User Acceptance Testing (UAT)

Date	29/09/2026
Team ID	SWTID-2026-2908
Project Name	FitSense – AI-Powered Fitness Recommendation System
Maximum Marks	[Enter Maximum Marks]

Project Overview

Project Name: FitSense – AI-Powered Fitness Recommendation System

Project Description: FitSense is an AI-powered fitness recommendation system that helps users manage workout records and receive personalized fitness recommendations based on their age, fitness goal, experience level, and workout history.

Project Version: Version 1.0

Testing Period: 29/09/2026 to [End Date]

Testing Scope

Features and Functionalities to be Tested:

• User registration
• Secure user login
• JWT authentication
• Adding workout records
• Viewing workout history
• Searching workouts
• Updating workout records
• Deleting workout records
• AI-generated workout recommendations
• Personalized fitness insights
• Workout plan generation
• Performance analysis
• Improvement suggestions
• Safety recommendations
• JSON API responses
• MongoDB data storage

User Stories / Requirements to be Tested:

1. As a user, I want to create an account so that I can use FitSense.
2. As a user, I want to log in securely.
3. As a user, I want to add my workout details.
4. As a user, I want to view my previous workouts.
5. As a user, I want to search my workout history.
6. As a user, I want to update or delete workout records.
7. As a user, I want personalized workout recommendations.
8. As a user, I want fitness insights based on my workout statistics.

Testing Environment

URL/Location: [Enter API/Application URL or Localhost Location]

Credentials: [Enter Username/Password if required]

Technology Environment: Node.js, Express.js, MongoDB, Mongoose, Google Gemini AI, JWT, bcrypt.js, Postman, REST API, JSON, MVC Architecture.

Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Actual Result	Pass/Fail
TC-001	User Registration	1. Open registration API 2. Enter name, email and password 3. Submit request	User account should be created successfully	[Record Result]	[Pass/Fail]
TC-002	User Login	1. Enter registered email 2. Enter password 3. Submit login request	User should be logged in and receive a JWT token	[Record Result]	[Pass/Fail]
TC-003	Invalid Login	1. Enter incorrect email/password 2. Submit request	System should reject login and return an error message	[Record Result]	[Pass/Fail]
TC-004	Add Workout	1. Login 2. Enter workout name, category, duration and calories 3. Submit	Workout should be stored successfully in MongoDB	[Record Result]	[Pass/Fail]
TC-005	View Workout History	1. Login 2. Request workout history	User's workout records should be displayed	[Record Result]	[Pass/Fail]
TC-006	Search Workout	1. Enter workout search keyword 2. Submit request	Matching workout records should be displayed	[Record Result]	[Pass/Fail]
TC-007	Update Workout	1. Select an existing workout 2. Change workout details 3. Submit update	Workout information should be updated successfully	[Record Result]	[Pass/Fail]
TC-008	Delete Workout	1. Select workout record 2. Send delete request	Selected workout should be deleted	[Record Result]	[Pass/Fail]
TC-009	AI Workout Recommendation	1. Enter age 2. Select fitness goal and experience 3. Provide workout statistics 4. Request recommendation	Gemini AI should generate a personalized workout plan	[Record Result]	[Pass/Fail]
TC-010	Fitness Insights	1. Provide workout history/statistics 2. Request fitness analysis	System should provide performance analysis and improvement suggestions	[Record Result]	[Pass/Fail]
TC-011	Safety Recommendation	1. Provide user fitness information 2. Request AI recommendation	System should provide relevant training and safety recommendations	[Record Result]	[Pass/Fail]
TC-012	Unauthorized Access	1. Send protected API request without JWT token	System should deny access and return an authentication error	[Record Result]	[Pass/Fail]

Bug Tracking

Bug ID	Bug Description	Steps to Reproduce	Severity	Status	Additional Feedback
BG-001	[Describe issue if found]	1. [Step 1] 2. [Step 2] 3. [Step 3]	[Low/Medium/High]	[Open/In Progress/Closed]	[Additional comments]
BG-002	[Describe issue if found]	1. [Step 1] 2. [Step 2] 3. [Step 3]	[Low/Medium/High]	[Open/In Progress/Closed]	[Additional comments]

Team Details

Team ID: SWTID-2026-2908
Team Size: 4
Team Leader: ELIZABETH MARY ROBERTS
Team Members: KOWSALYA N, NEDHA THASEF H, RAFATHUNISSA S

Sign-off

Tester Name: [Name of Tester]

Date: 29/09/2026

Signature: [Tester’s Signature]

Notes:
• All test cases should cover both positive and negative scenarios.
• Testers should provide detailed feedback and suggestions.
• Any bugs should include severity, status, and reproduction steps.
• Sign-off should be obtained from the project manager and product owner before deployment.